

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUV♥
HYDROCARBONS

LECTURE- 05

CHEMISTRY

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Topics To Covered

Alkanes ✓
Alkenes
Alkynes
Aromatic hydrocarbon



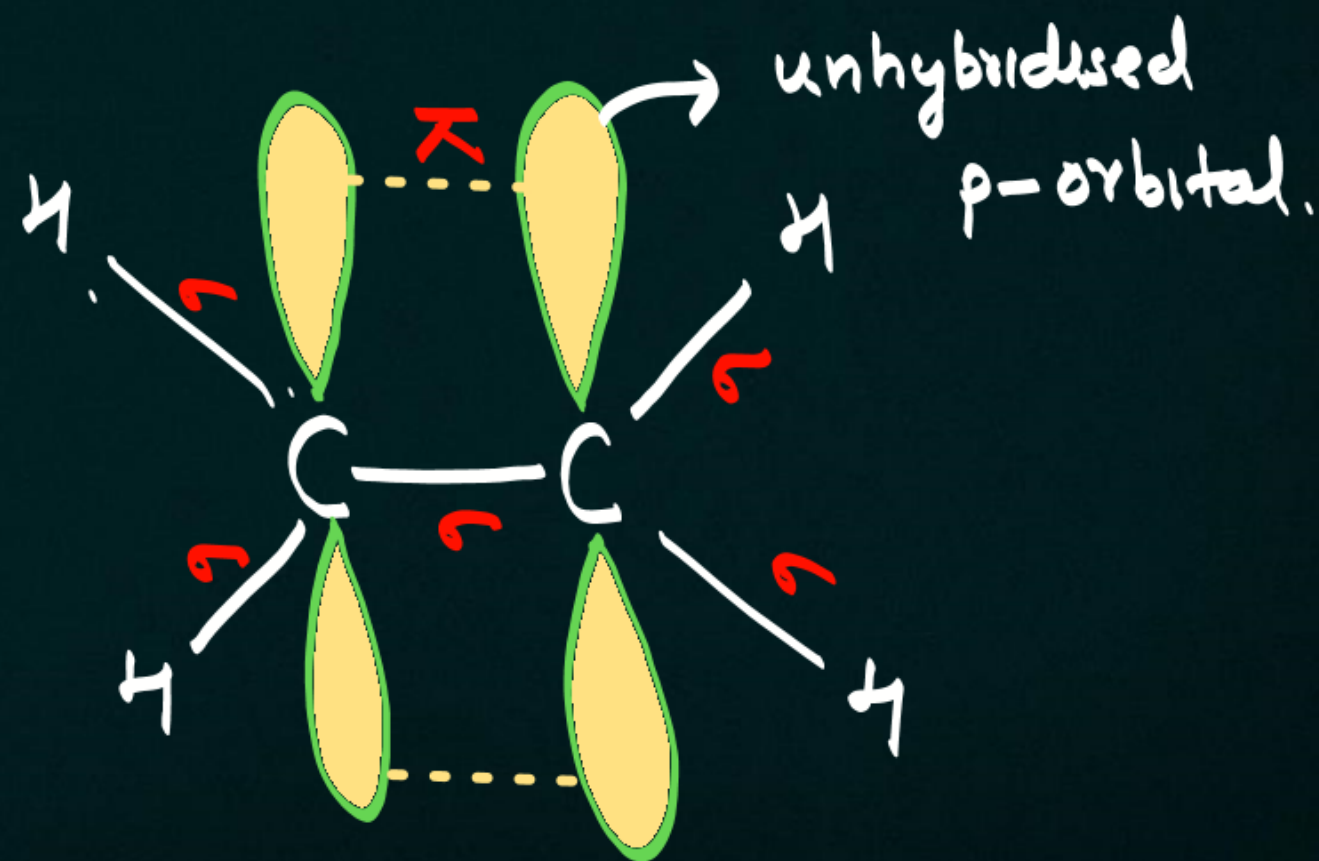
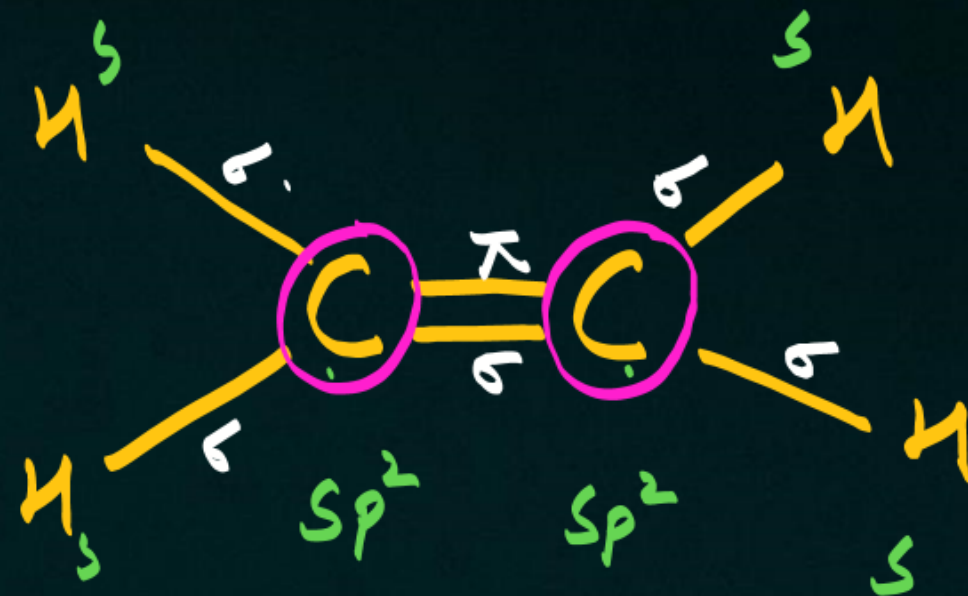
- 1 Structure of Alkenes
- 2 Geometrical Isomerism of Alkenes
- 3 Preparations of Alkenes

A decorative, ornate frame made of a golden-yellow material with a metallic sheen. It features elegant scrollwork at the top, bottom, and sides, enclosing the text.

PART – 2:
Alkenes (C_nH_{2n})

STRUCTURE OF ALKENE.

- Let us take the example of ethene.
- Each carbon has 3 σ -bonds and one π -bond that means each carbon has sp^2 hybridisation.
- Out of three sigma bonds, C-C sigma bond is formed by the overlap of sp^2-sp^2 hybrid orbital and two C-H bond are formed by the overlap of $sp^2(C) + s(H)$ orbitals.
- And C-C π bond is formed by the sideways overlap of unhybridised p-orbital perpendicular to the plane.
- Due to this π -bond alkene is e^- -rich & shows electrophilic reactions.

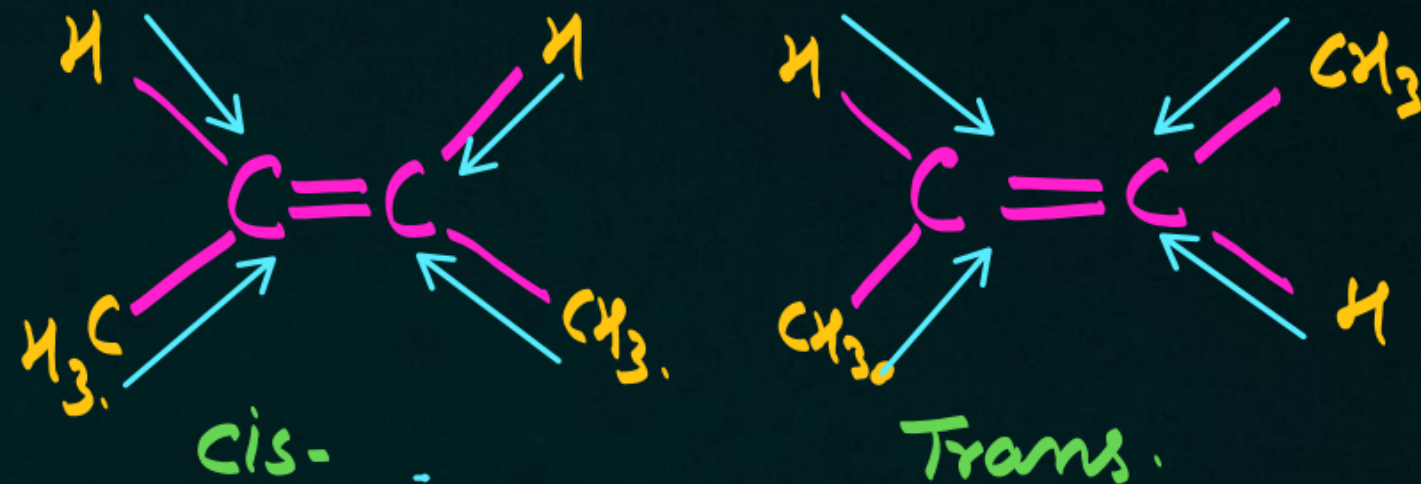


GEOMETRICAL ISOMERISM.

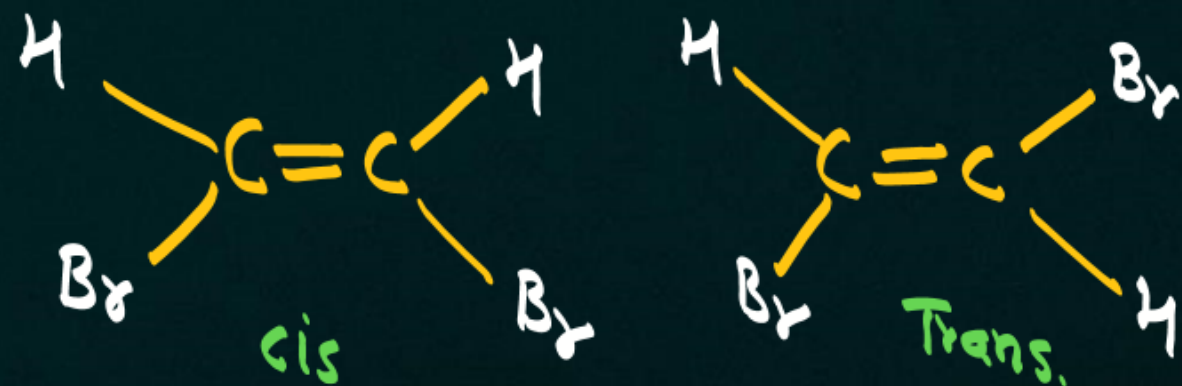
Molecules having same molecular formula but different arrangement of atoms or groups in space around the double bond, are called Geometrical Isomers & phenomena is known as Geometrical isomerism.

- Type of isomer have same groups or atoms, are present on same side of double bond, are called "Cis-isomer".
- Type of isomer have same groups or atoms, present on opposite side of double bond are called "Trans-isomer".

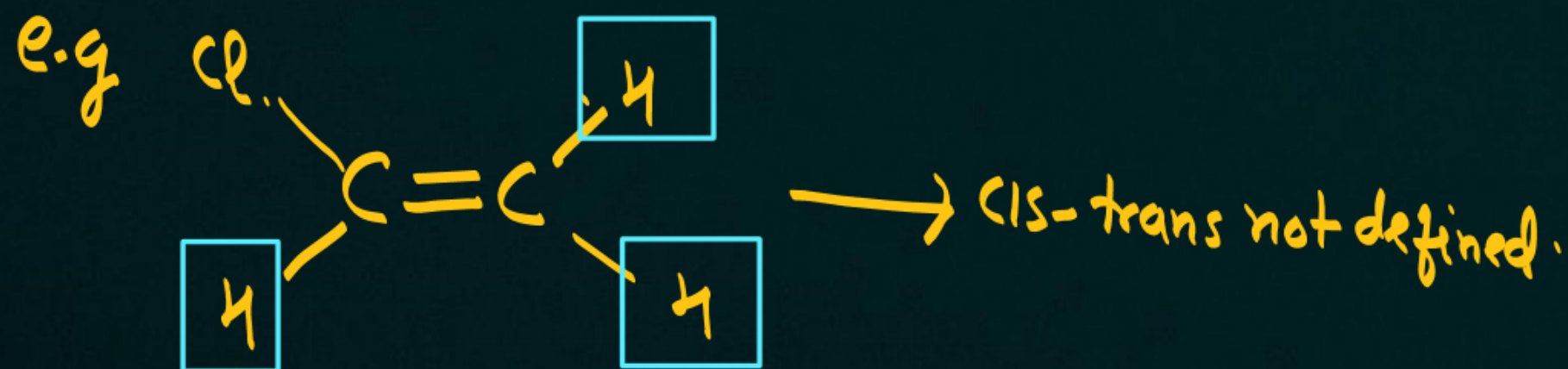
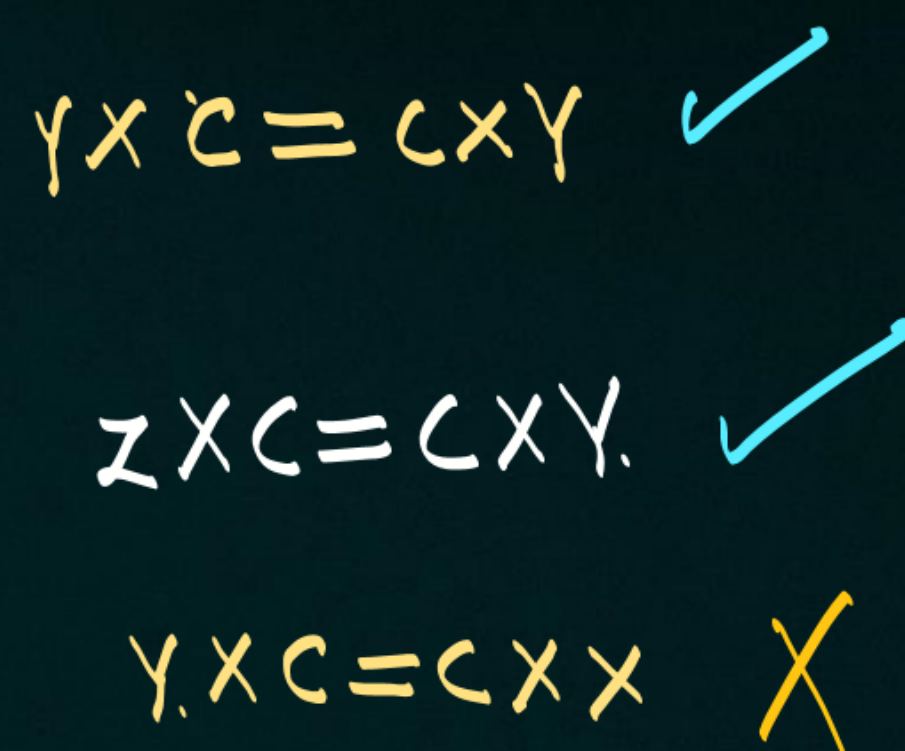
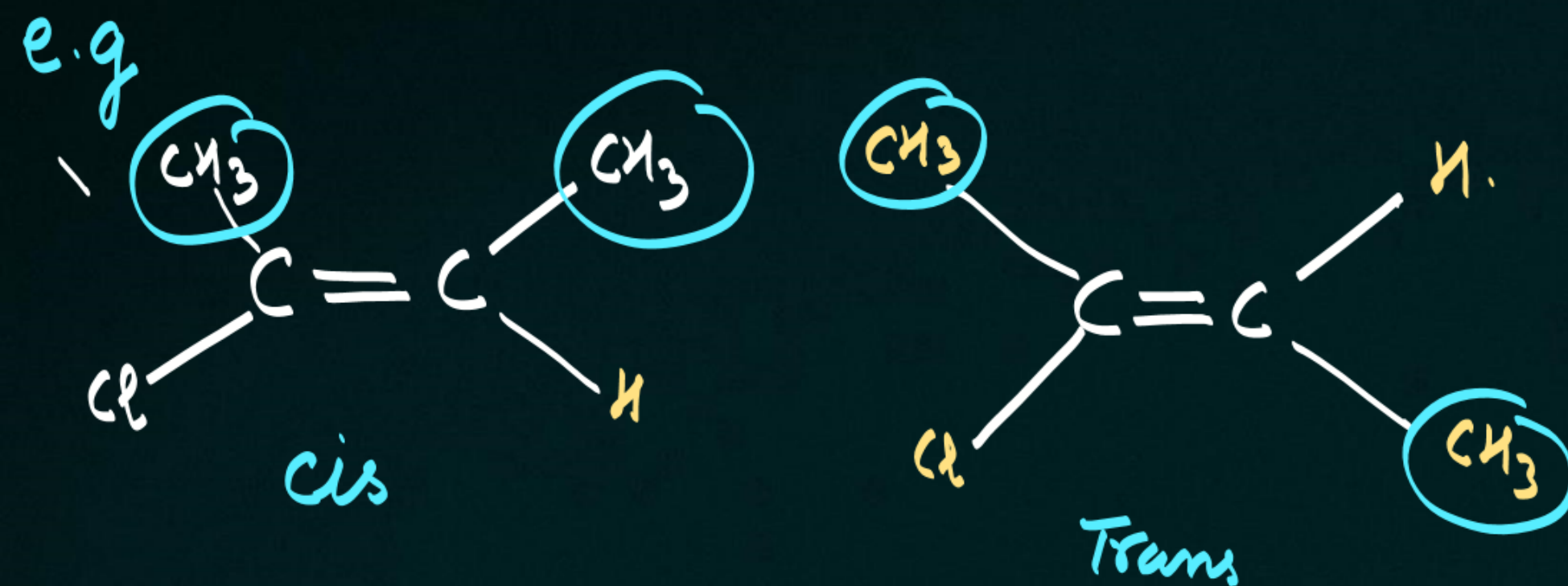
e.g But-2-ene ($\text{CH}_3\text{-CH=CH-CH}_3$)



e.g. 1,2-Dibromoethene. (Br-CH=CH-Br).



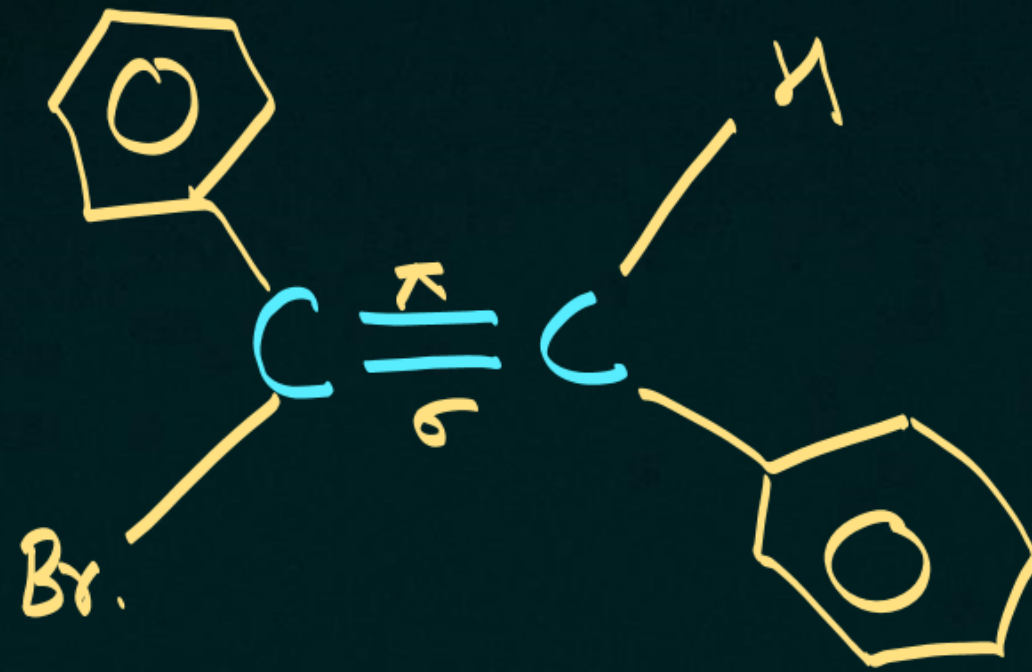
- Cis-isomer has more dipole moment (polar) than Trans-isomer.

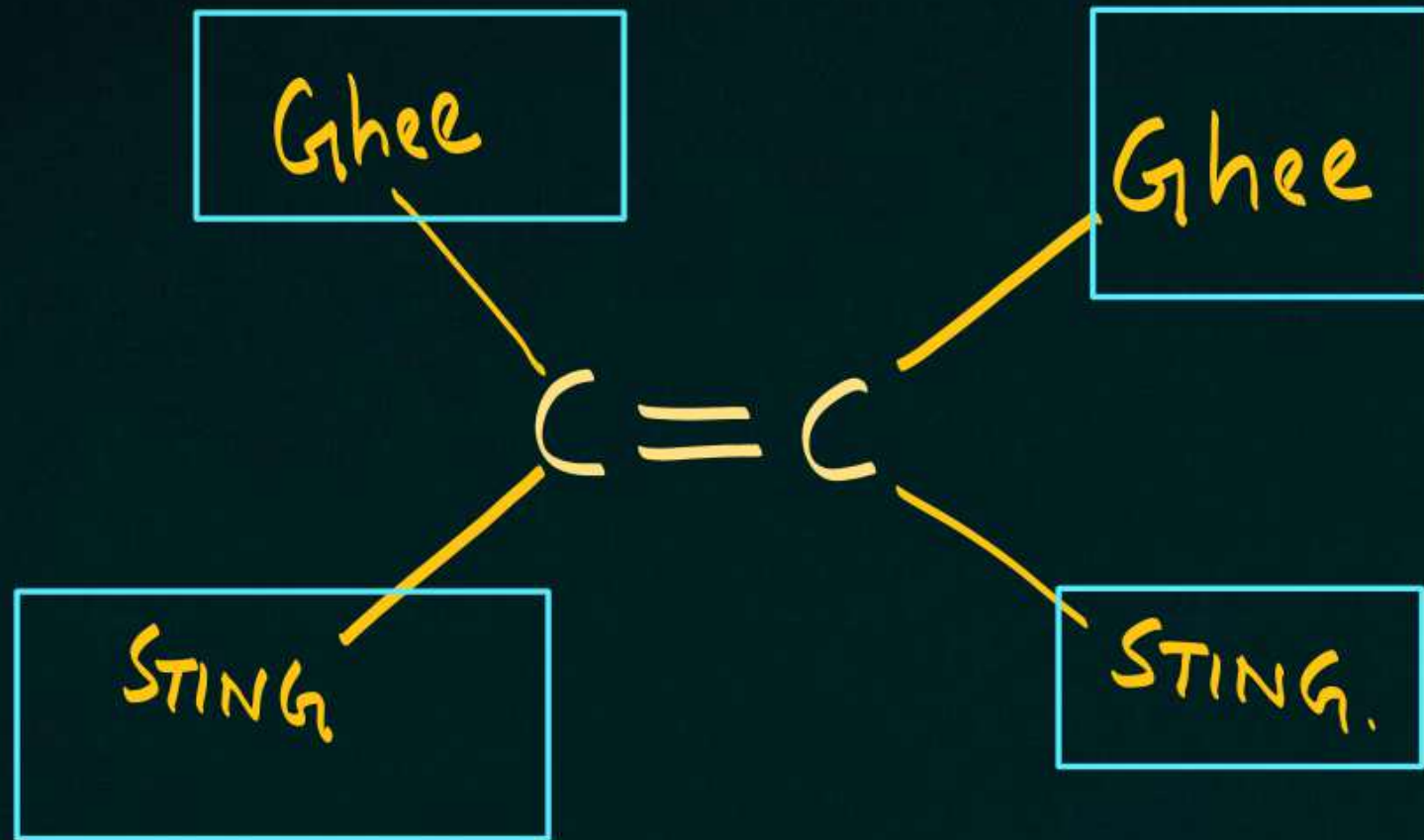


Q. Is this molecule.
Trans

a) Yes

b) No.





CIS
OR
TRANS.

EXAMPLE

Which of the following compounds will show *cis-trans* isomerism?

- (i) $(\text{CH}_3)_2\text{C} = \text{CH} - \text{C}_2\text{H}_5$
- (ii) $\text{CH}_2 = \text{CBr}_2$
- (iii) $\text{C}_6\text{H}_5\text{CH} = \text{CH} - \text{CH}_3$
- (iv) $\text{CH}_3\text{CH} = \text{CClCH}_3$

PREPARATION OF ALKENE.

MOP of alkanes
i) From Hydrogenation
ii) Alkyl halide $\rightarrow$ Red Wittig Rxn.
iii) COOH $\rightarrow$ Decarboxylation
 $\rightarrow$ Kolbe electrolysis.



FROM ALKYNE.

FROM ALKYL HALIDE

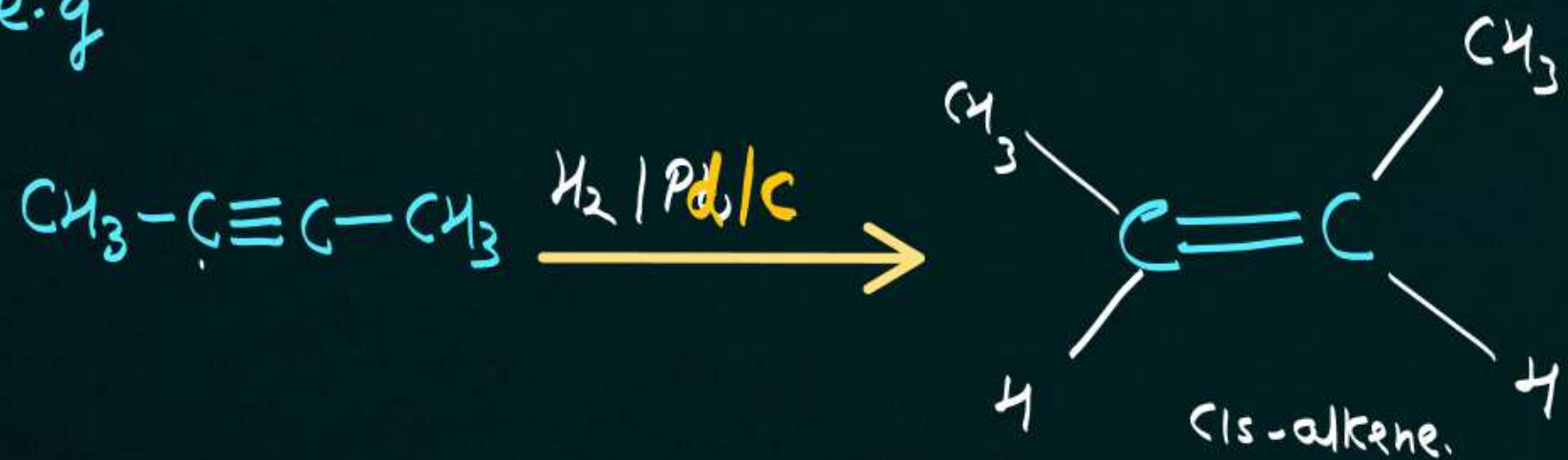
FROM VICINAL
DIALKIDES.

From Dehydration
of Alcohol

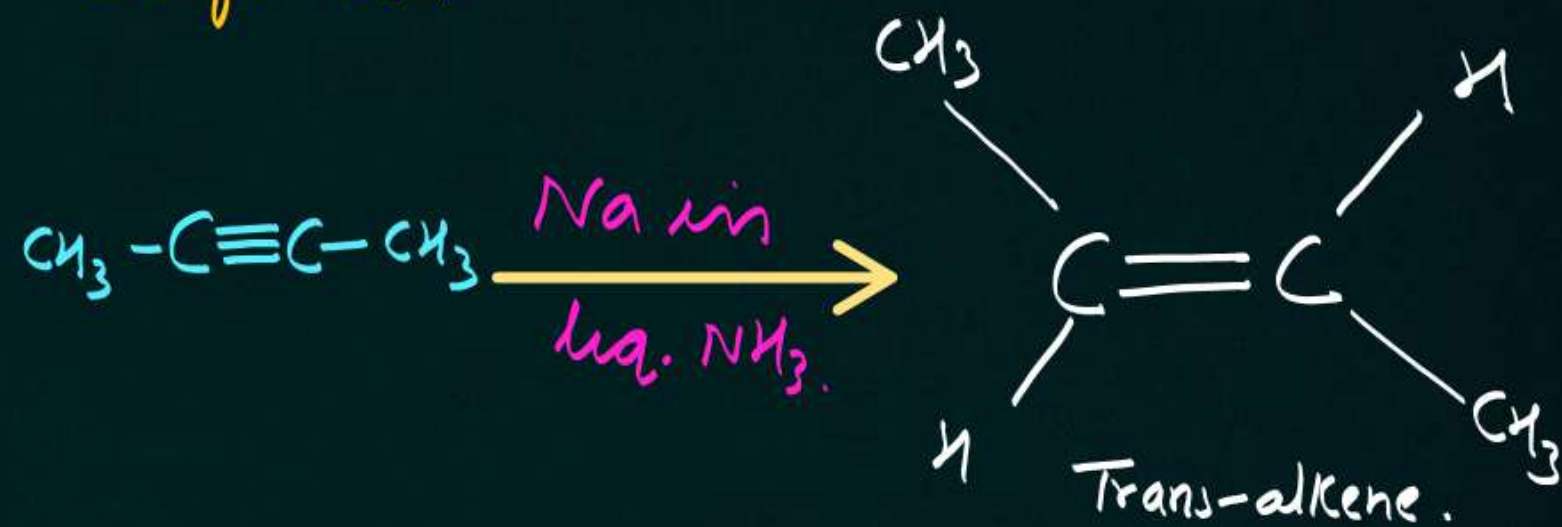
1) FROM ALKYNE.

- ALKYNES on treatment with H_2 in presence of metal catalyst (Ni, Pt, Pd etc) ^{or Carbon.} gives hydrogenation rxn & produces cis-alkene.

e.g

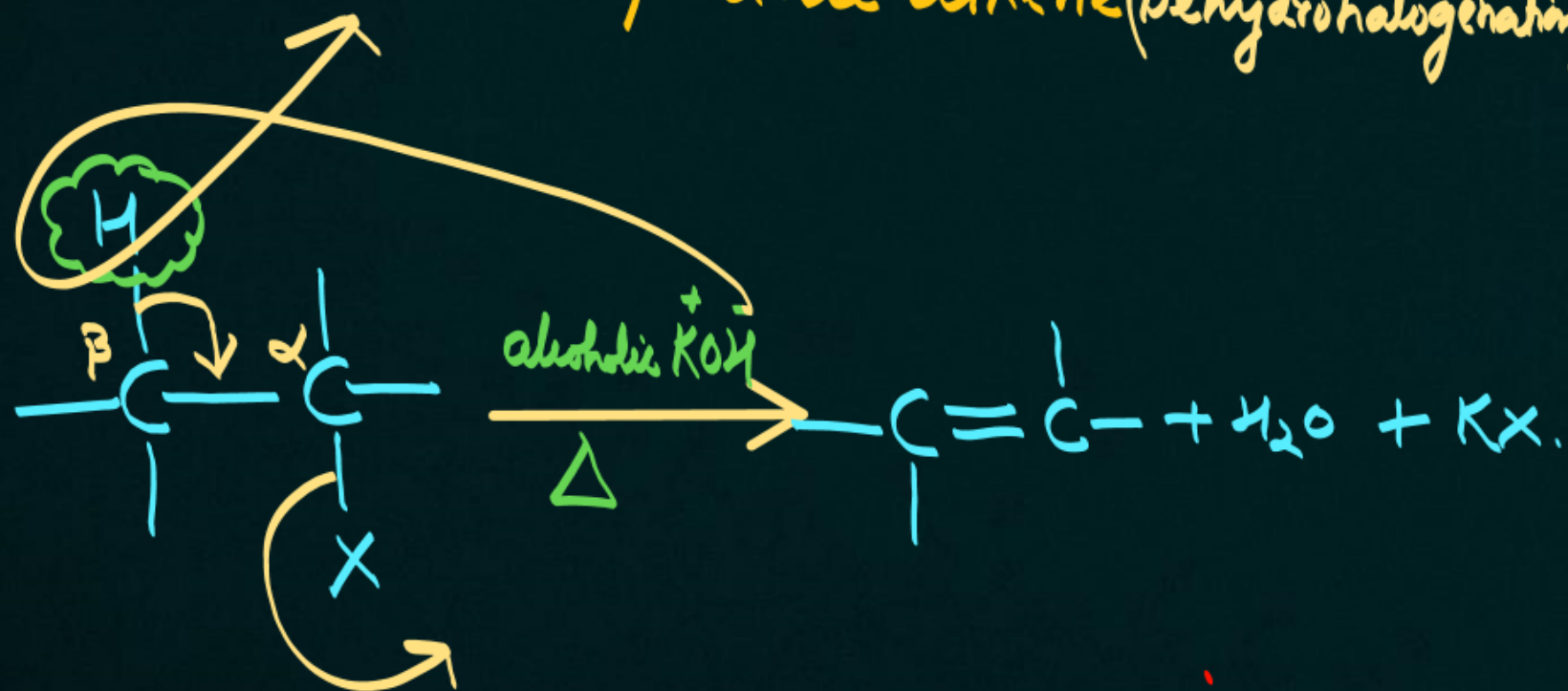


- But if alkyne is allowed to react with Na in liq. NH_3 then trans-alkene is formed.

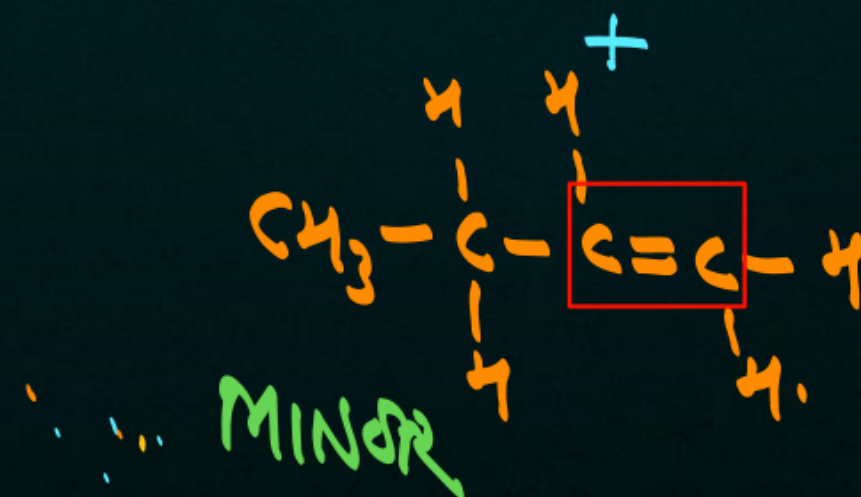


2) FROM ALKYL HALIDE. $R-X$

Alkyl halides having β -Hydrogen when heated with alcoholic KOH , undergo β -elimination to produce alkene (dehydrohalogenation).



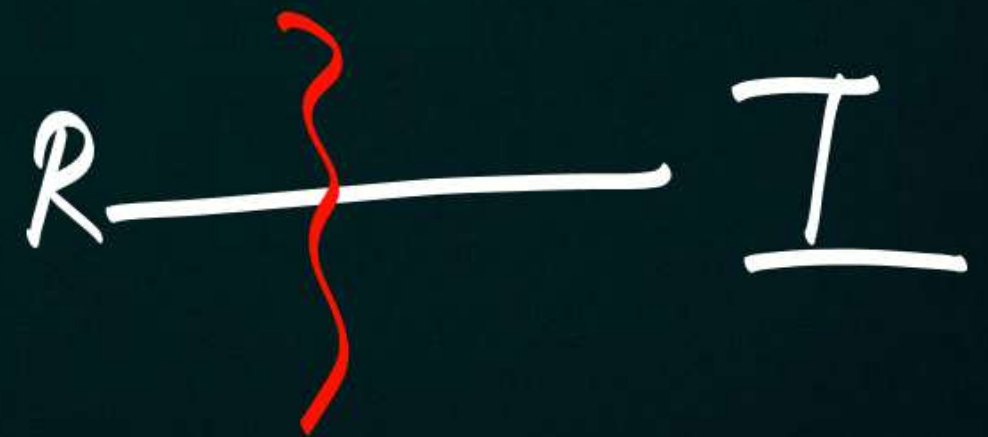
In case of alkyl halides having more than one type of β -H, then alkene is formed according to "Saytzeff Rule" which states that "in case of β -elimination preferred product is that, which has more no. of alkyl groups on double bond".



ORDER OF REACTIVITY OF ALKYL HALIDES.



In case of halogens



REVISE KARNA.

Thank
You

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